

# STEEL

Self  
confidence

Achieve  
Daily Target

Multiple  
revision

To The Point-ByDhyanPal(ESE'17AIR-179,GATE'18AIR-93,GATE'16AIR-145)



## IS rolled steel beam section :- (4) series

| Section                      | classification                                                                                                                         |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Beam                         | ISJB - Ind. std. junior Beam<br>ISLB → light weight beam<br>ISMB → medium weight beam<br>ISWB → wide flange beam                       |
| column or Heavy weight beams | ISSC → column sections<br>ISHB → Heavy weight beam                                                                                     |
| channels                     | ISJC → junior channels<br>ISLC → light weight channel<br>ISMC → medium weight channel<br>ISMCP → medium weight parallel-flange channel |
| Angles                       | ISA → equal angles<br>ISA → unequal angles<br>ISBA → Bulb angles                                                                       |

## various types of rolled structural steel section

(i) Rolled steel I section

(ii) channel "

(iii) Angle "

(iv) T "

(v) Tube "

(vi) bars "

(vii) flats "

(viii) plates "

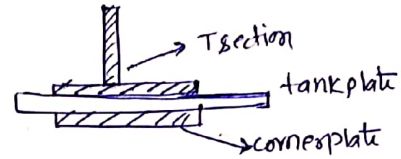
(ix) sheets "

(x) strips "

## T-section Application :-

(i) used to transmit bracket loads to column.

(ii) used with flat strips to connect plates in the steel ~~rectangular~~ rectangular tanks.



## Angle section :- uses :-

- designed for resisting axial force (comp & tension) and transverse forces as purlins

- they may be used as connecting elements to connect structural elements like sheets/plates or to form a built up section.

## Bulb section :-

- provide better plate stiffening
- used in ship building

- when the stress is under extreme stress and starts to buckle, this shape is highly resistant and increases the longevity of the structure.

## main advantage of steel member :-

(i) High strength to weight ratio

(ii) Gas & water tightness

(iii) Longer Life

(iv) No sudden failure

(v) Economy in transportation & Handling

(vi) Termite proof & rot proof.

(vii) formwork

Steel density  
7850 kg/m<sup>3</sup>



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## Rolled steel beams :

- ① mainly used to resist bending stress when they are designed as beam.
- ② are used as independent section to resist compressive stress when they are designed for column / strut.
- ③ are used as independent section to resist tensile stress when they are designed as tie.

design wind pressure  
(IS 875 part-3)

$$P_z = 0.6 V_z^2$$

(N/m<sup>2</sup>)      (m/s)

$$V_z \text{ design wind velocity} = V_0 \times \{k_1 \times k_2 \times k_3\}$$

$k_1$  = Risk coefficient or Probability factor

$k_2$  = Height, terrain, str. size factor

$k_3$  = Topography factor.

## Method of Design of Steel framework :-

### Simple design

- Based on elastic theory
- assume pin joint (no moment transfer)
- all connection of beams, girders, trusses are virtually flexible.
- most uneconomical method

### Semi-rigid Design

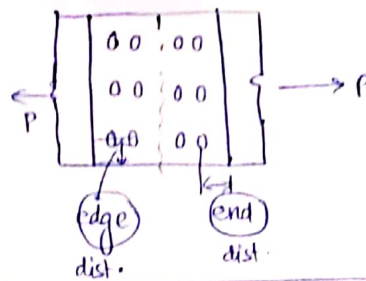
- ensure partial fixity is available at support
- This method permits a reduction in max. BM in beam suitably connected to support due to partial transfer of moment to another connected member.
- economical than simple design

### Rigid Design

- assumption → end connections are fully rigid and capable of transmitting moment & shears
- frames sufficient rigid to hold virtually unchanged original angles b/w such members and member they connect.
- It is used in convenient cases and gives economy in the weight of steel and saves construction cost



|                             |                                    |         |
|-----------------------------|------------------------------------|---------|
| axial compression/tension   | $\frac{f_y}{1.1}$                  | } $f_y$ |
| bending compression/tension | $\frac{f_y}{1.1}$                  |         |
| structural steel            | $\frac{f_y}{\sqrt{3} \times 1.1}$  | } $f_y$ |
| bolt                        | $\frac{f_u}{\sqrt{3} \times 1.25}$ |         |



design strength of bolted connection  $\Rightarrow \min$  {

- strength of bolt in shear ( $V_{dsb}$ )
- strength of bolt in bearing ( $V_{dwb}$ )
- strength of bolt in tension ( $T_{db}$ ) (if exist)

min edge/end dist  $\begin{cases} \text{handflamed out edges \& sheared} = 1.7\phi \\ \text{machined flamecut sawn, plane edges} = 1.5\phi \end{cases}$

max. edge distance =  $12t\epsilon$  ( $\epsilon = \sqrt{\frac{250}{f_y}}$ )  $\begin{cases} t \rightarrow \text{thinner} \\ \rightarrow \text{out of plate} \end{cases}$   
(unstiffened part)  
note :- This will not apply to fasteners interconnecting the components of back to back tension member.

design shear strength of bolt or shear capacity of bolt  $V_{dsb} = \frac{f_{ub}}{\sqrt{3} \times 1.25} [n_s A_{nb} + n_t A_{sb}]$

where member exposed to corrosion  $\rightarrow$   
max edge dist  $\Rightarrow 40 + 4t$

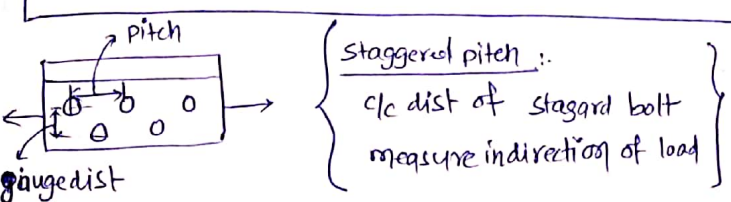
$n_s \rightarrow$  no. of shear plane with thread intercepting plane  
 $A_{nb} \Rightarrow$  net shear area of bolt at thread =  $0.78 A_{sb}$   
 $= 0.78 \times \frac{\pi}{4} d^2$   
 $A_{sb} \Rightarrow$  nominal shank area of bolt

min end/edge dist. required 'or' Bolt holes should not placed too near to the edge Reasons  $\rightarrow$

note:- Reduction factors for shear capacity of Bolt :-

- ① The failure of Plate in tension may take place
- ② The steel of the plate opposite to the hole may bulge out and may crack

|                          |                                                                                                     |
|--------------------------|-----------------------------------------------------------------------------------------------------|
| long joint ( $\beta_j$ ) | Bolt ( $L_j > 15d$ )<br>$\beta = 1.075 - (0.5\%) \frac{L_j}{d}$ $\frac{0.75}{75} \leq \beta \leq 1$ |
|                          | weld ( $L_j > 150t$ ) $\rightarrow$ Throat thickness<br>$\beta = 1.2 - 0.2 \frac{L_j}{150t}$        |
| lap length               | $\beta = \frac{8d}{3d + L_g}$ ( $L_g > 5d$ )                                                        |
| packing plates           | ( $t_{pk} > 6\text{mm}$ ) $\beta = 1 - 125 \times 10^{-4} t_{pk}$                                   |



Bearing capacity of bolt  $= \frac{2.5 k_b d t f_u}{1.25}$

- min pitch required ?
- ① to prevent failure of member b/w 2 consecutive rivets.
  - ② to permit efficient installation of rivet
  - ③ to provide adequate resistance to tearout of bolt/rivet

$k_b = \min \left\{ \frac{e}{3d_0}, \frac{p}{3d_0} - 0.25, \frac{f_u}{f_{ub}}, 1.0 \right\}$   
 $\begin{cases} e \rightarrow \text{end} \\ d_0 \rightarrow \text{bolt hole} \\ p \rightarrow \text{pitch} \end{cases}$   
 $\begin{cases} f_u \rightarrow \text{plate} \\ f_{ub} \rightarrow \text{bolt} \end{cases}$

min pitch  $\rightarrow 2.5\phi$

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$(327, 300) \rightarrow$  not exposed to weather

Note:- tack rivets are used when a member (consist of 2 elements) section, which is suppose to act as single unit then they are connected by tack bolts.

- compression member :- where force are transferred through Butting face

The distance b/w the centres of any consecutive fasteners in a line adjacent and parallel to an edge of an outside plate

special note :- if staggered fastener  $\Rightarrow$   
at equal interval &  $g \geq 75\text{mm}$   
then spacing\* increase by 50%.

mm

LSm

bottom hole (clearance)

| $\phi(d)$ | $\phi'(d_0)$ |
|-----------|--------------|
| $\leq 14$ | +1           |
| $> 24$    | +3           |
| (16-24)   | +2           |

| wsm       |              |
|-----------|--------------|
| river d   | d1 (cleaner) |
| $\leq 25$ | +1.5         |
| $> 25$    | +2           |

Bolt subjected to  $\left\{ \begin{array}{l} \text{Shear} \\ + \\ \text{Tension} \end{array} \right.$   $\left[ \left( \frac{V_u}{V_d} \right)^2 + \left( \frac{T_u}{T_d} \right)^2 \leq 1 \right]$

Block shear strength : combinations of yielding + rupture (tension)

- 

① Shear yielding + Tensile rupture

$$\left( \frac{A_{gv} \times f_y}{\sqrt{3} \times 1.1} \right) + \left( A_{tn} \times \frac{0.9 f_u}{1.25} \right)$$

$$\text{Block shear strength} = \min \{ (1), (2) \}$$

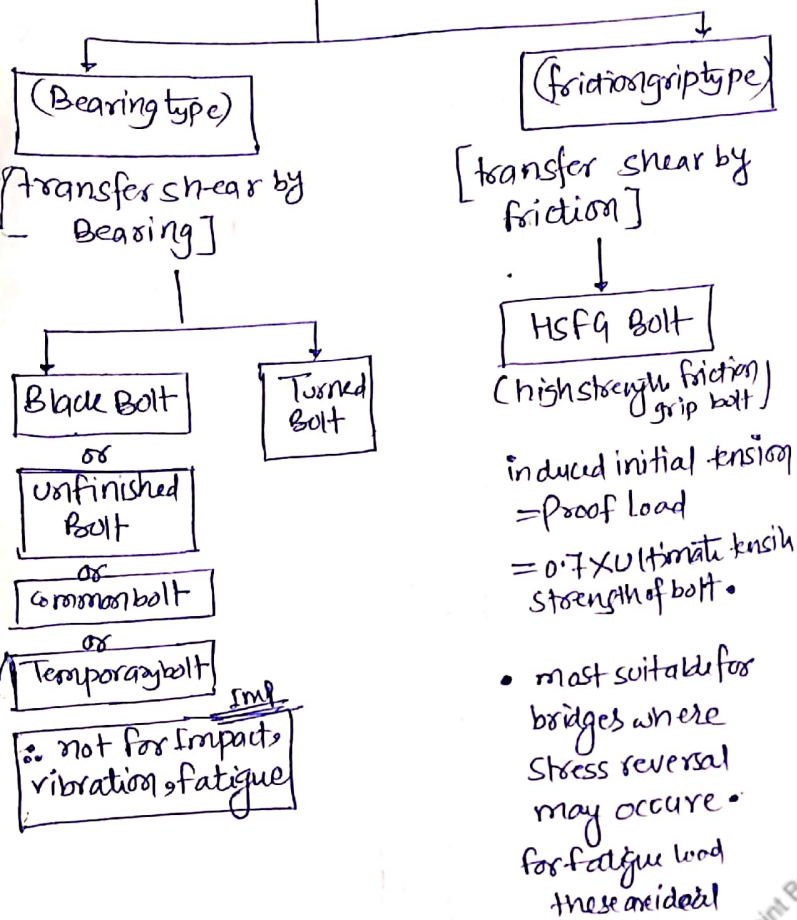
Ex.   $d' = 18$   $\left\{ \begin{array}{l} \text{Avg} = 150 \times 8 \\ \text{Ava} = [150 - (1.5 \times 18)] 8 \\ \text{Alg} = 35 \times 8 \\ \text{Abn} = (35 - 0.5 \times 18) \times 8 \end{array} \right.$

Prying force :- HSFG bolts  $\rightarrow$  subjected to tensile force  
then additional forces are considered due to flexibility  
of connected part



[on basis of Load Transfer]

Bolt type



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## Assumption in design of Rivetted connection :-

- 1- Rivets are assumed to be stressed equally.
- 2- The rivet hole is assumed to be completely filled by rivet.
- 3- friction b/w plate  $\longrightarrow$  neglected
- 4- Shear stress is assumed to be uniformly distributed over the gross cross sectional area of rivet.
- 5- stress in a plate is assumed to be uniform
- 6- Bending of rivet  $\longrightarrow$  neglected.
- 7- Bending stress is uniform b/w plate & rivet.

| WSM            | clearance |
|----------------|-----------|
| $\phi \leq 25$ | +1.5mm    |
| $\phi > 25$    | +2 mm     |

unwin's formula

$$\phi = 6.01 \sqrt{t \text{ min (mm)}}$$

( $t \text{ min} > 8$ )

( $t < 8$ )

$$\sigma_{scdt} = \frac{\pi}{4} d^2 \sigma_{sc}$$

$\downarrow$  shear                       $\downarrow$  bearing

IS: 800 : 1984 :

| permissible stress          | value      |
|-----------------------------|------------|
| avg. shear stress           | $0.40 f_y$ |
| max. shear stress           | $0.45 f_y$ |
| axial comp/tensile stress   | $0.60 f_y$ |
| Bending comp/tensile stress | $0.66 f_y$ |
| Bearing stress              | $0.75 f_y$ |
| combined bearing & bending  | $0.90 f_y$ |

Permissible stress in shop rivets :-

| type of rivet | shear | bearing | axial tension |
|---------------|-------|---------|---------------|
| Power driven  | 100   | 300     | 100           |
| Hand driven   | 80    | 250     | 80            |

note:- ① field rivets :-  $-10\%$  these values

② if WL or EQ load is considered then

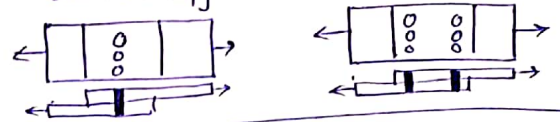
$+25\%$  above values.

## Types of joints :-

### Lap joint

when 2 members to be connected are overlapped & connected together

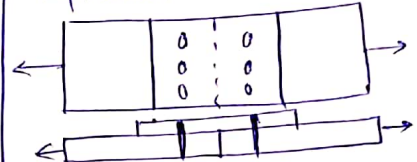
(i) single bolted lap joint (ii) double bolted lap joint



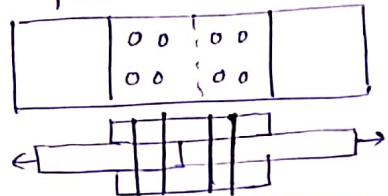
### Butt joint

2 members are placed end to end and are joined by cover plates.

① Single cover butt joint :- when 1 cover plate is provided on one side.



② Double cover butt joint :- when 2 cover plates are provided.

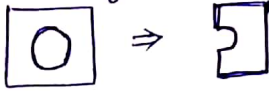


| Joint               | n. %  |
|---------------------|-------|
| Lap single riveted  | 50-60 |
| double "            | 60-70 |
| Tripple "           | 72-80 |
| Butt single riveted | 55-60 |
| double riveted      | 76-84 |
| Tripple riveted     | 80-88 |

## Various modes of failure of riveted connection :-

### (A) failure of plate

① By failure of plate b/w rivet hole & edge



- due to less/insufficient edge dist.
- ⇒ to avoid this failure provide min edge dist.

② Tearing of plate b/w rivet holes :-

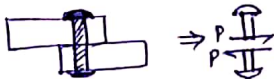


Tearing strength of plate  

$$= (b - nd')t \sigma_t$$
 where  $n$  = no. of rivets

### (B) failure of rivet

① failure of rivet in shear :-



Strength of rivet in shear  $= \sigma_{\text{shear}} \times \frac{\pi}{4} d^2$

② failure of rivet in bearing :-

Strength of rivet in bearing



$$= \sigma_{br} \times d \times t$$

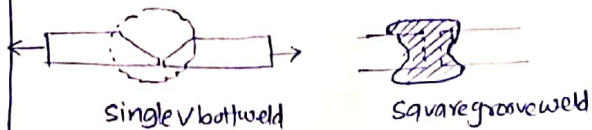
↓  
thickness of thinner plate



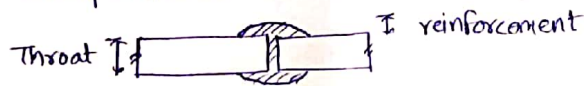
## Butt weld (Groove weld)

• Provided when member to be jointed are in one plane (i.e. But joint)

• Butt weld refers to the beads that are deposited in a groove b/w 2 members to be joined



• Size of Butt weld = throat dimension (effective throat thickness)



effective Throat thickness ( $t_t$ )

double V  
in complete penetration  
(thickness of thinner member)

(single V groove weld)  
in incomplete penetration  
( $5/8$  of thinner thickness)

note:- reinforcement  $\rightarrow$  extra weld metal which makes the throat thickness at least 10% greater than thickness of welded material  
fn  $\rightarrow$  to increase  $\eta$  of joint.

- reinforcement  $\nless 3$  mm
- In calculation neglect reinforcement.

axial Strength of Butt weld :- govern by yielding (in comp. or tension)

$$\Rightarrow \frac{(l_{eff} \times t_t) \times f_y}{1.25} \rightarrow \text{for shop weld}$$

$$\Rightarrow \frac{(l_{eff} \times t_t) \times f_y}{1.50} \rightarrow \text{for field weld}$$

basically  $f_y \rightarrow \min \left\{ \begin{array}{l} \text{strength of weld} \\ f_{yw} \end{array} \right\}, \text{ yield strength of parent metal } f_y$

Shear strength of butt weld :- govern by yielding

$$\Rightarrow \frac{(l_{eff} \times t_t) \times f_y}{\sqrt{3} \times 1.25} \rightarrow \text{shop weld}$$

$$1.50 \rightarrow \text{field weld}$$

effective length of Butt weld = length of full size weld

• min length of Butt weld =  $4 \times \text{size of weld}$   
 $= 4S$

intermittent Butt welding :-

min effective length =  $4 \times S$

$\&$  spacing  $\nless 16t_{min}$  (thinner)

note :- The intermittent weld shall not be used in position subjected to dynamic, repetitive and alternating stresses.

Butt weld shall be treated as parent metal with a thickness equal to throat thickness.  
stress  $\nless$  permitted in parent metal.

check for combination of stress in Butt weld need not to be carried out if  $\rightarrow$

(i) Butt welds are axial loaded.

(ii) in single & double bevel weld :-

[ sum of normal + shear stress  $\nless$  design normal stress ]

[ shear stress  $\nless$  50% of design shear stress ]

Butt weld :-

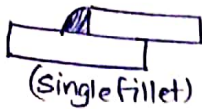
combined bearing, bending, shear :-

$$f_{eq} = \sqrt{(f_{br})^2 + (f_b)^2 + (f_{br}f_b) + 3q^2}$$



## fillet weld :- (shear stress criteria)

- Provided when 2 members to be jointed are in different plane and Lap joint

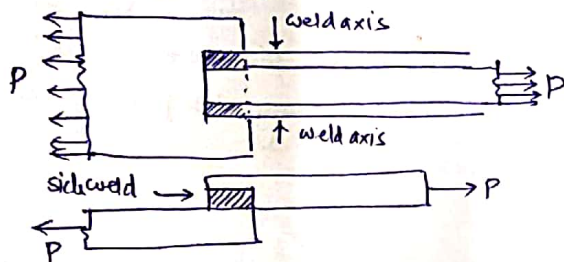


- a fillet weld joins 2 surfaces at approximate right angle to each other.

Side fillet weld :- when member with side weld is loaded, the load axis is parallel to the weld axis.

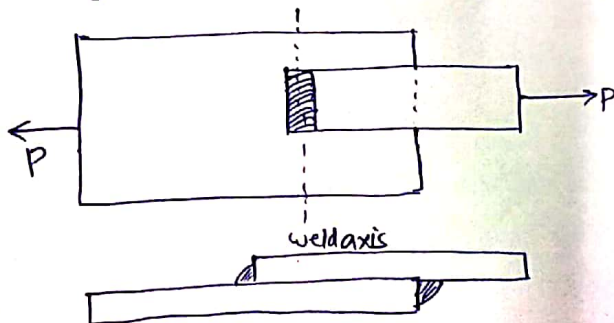
- weld is subjected to shear
- weld shear strength is limited to just about half the weld metal tensile strength

- ductility is high in side weld

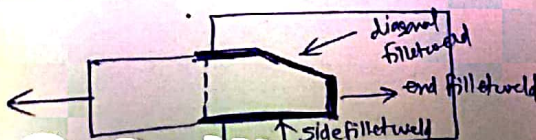


end fillet weld :- when a member with end weld is loaded, the load axis is perpendicular to weld axis.

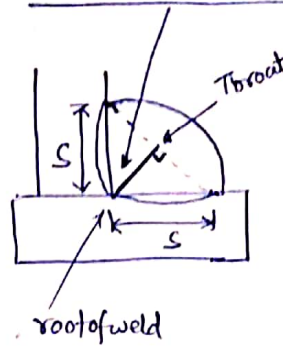
- when a connection with end fillet is loaded in tension, The weld high strength & the strength developed in weld = value of weld metal.



note :- end fillet weld stronger than side weld. But for calculation & analysis both are same.



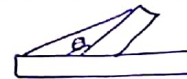
effective throat thickness ( $t$ ) =  $kS$  ← size of weld



- $t_{min} = 3 \text{ mm}$
- $t_{max} \rightarrow 0.7 \times \text{thinner thickness}$
- $t_{max} \rightarrow 1 \times \text{thickness of thinner member}$  (special case)

$S \rightarrow$  size of weld (length of smallest side of triangle length)  
 $t \rightarrow$  throat thickness (shortest perpendicular distance from root to hypotenuse)

$k \rightarrow$  depends on fusion faces angle



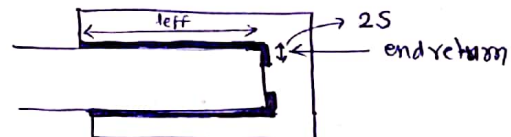
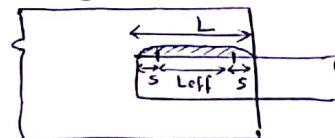
| $\theta$ degree | $k$  |
|-----------------|------|
| 60-90           | 0.70 |
| 91-100          | 0.65 |
| 101-106         | 0.60 |
| 107-113         | 0.55 |
| 114-120         | 0.50 |

\* fillet weld not recommended if  $\theta < 60$  &  $\theta > 120$

effective length of fillet weld :-

$$l_{eff} \leq 4S \leftarrow \left\{ \begin{array}{l} l_{eff} = L - 2S \\ \text{total length} \end{array} \right\} \left\{ \begin{array}{l} \text{or} \\ L = l_{eff} + 2S \end{array} \right.$$

- for which specified size & throat thickness of weld exist.
- in practice actual length of weld is made of the effective length + plus 2 times weld size



Intermittent fillet weld :- min effective length =  $\max(40, 4S)$

max. clear spacing :-  
 compression  $\rightarrow (12t, 200)$   
 tension  $\rightarrow (16t, 200)$

min gap in Lap joint  $\Rightarrow \max(40, 4t)$   
 ← thinner



min. size of weld :- (Based on thickness of thicker plate)  
 $(S)_{min}$

| thickness of thicker plate (mm) | size of weld mm (S)                |
|---------------------------------|------------------------------------|
| 0-10                            | 3                                  |
| 10-20                           | 5                                  |
| 20-32                           | 6                                  |
| 32-50                           | 8mm (1st run)<br>10mm (final size) |

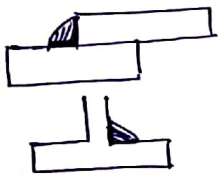
note: ① min size ?? to avoid risk of cracking in absence of Preheating.

② for thicker plate > 50mm special precaution like Preheating of plate will be taken.

max. size of weld :-  $\rightarrow$  based on thinner plate thickness

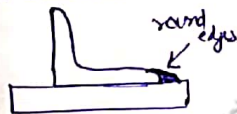
for square edges

$\Rightarrow$  thickness of thinner plate  $- 1.5$



for round edges

$\Rightarrow \frac{3}{4} \times$  nominal thickness of round edge



$\rightarrow$  Based on throat area.

Design shear strength of fillet weld :-

$$\Rightarrow (k_s \times t) \times f_u$$

$\sqrt{3} \times 1.25 \rightarrow$  shop  
 $1.50 \rightarrow$  field

$\rightarrow$  smaller of ultimate strength of weld & parent metal

Reduction factor for longitudinal weld :-

IF  $(L_j > 150t)$   $\beta = 1.2 - \frac{0.2 L_j}{150t} \leq 1$

$L_j \rightarrow$  length of joint or length of side fillet weld in direction of force.

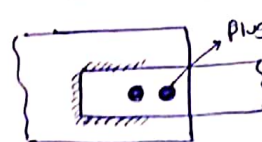
combined stress in fillet weld  $\begin{cases} \text{normal stress } (f_a) < \sigma_t \\ \text{shear stress } (q_s) \end{cases}$

$$f_{eq} = \sqrt{f_a^2 + 3q_s^2} \leq \frac{f_u}{\sqrt{3} \times 1.25 \rightarrow \text{shop} \quad 1.50 \rightarrow \text{field}}$$

no need to check combination of stress if

- ① side fillet weld joining cover plates & flange plates.
- ② fillet weld where normal stress + shear stress  $\neq \frac{f_u}{\sqrt{3} \times 1.25 \rightarrow \text{shop} \quad 1.50 \rightarrow \text{field}}$

Plug weld :-



$\rightarrow$  in which small holes are made in one plate and kept over another plate to be connected & then entire hole is filled with filler material

Slot weld :- used when overlapping length of weld is smaller than required weld length so to increase weld length are made



$\rightarrow$  in which a plate with circular hole is kept with another plate to be joined & then fillet weld is made along the periphery of hole.



**Ties** → steel members designed to carry axial tensile load (tension member)

tie beam in a truss is a horizontal beam connecting 2 rafters.

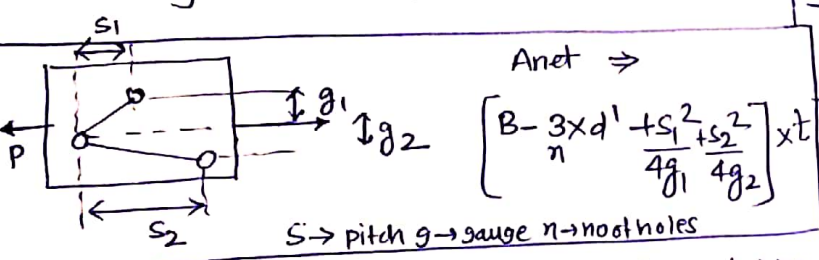
various forms of tension member :-

① wires → wire ropes are exclusively used for hoisting purpose

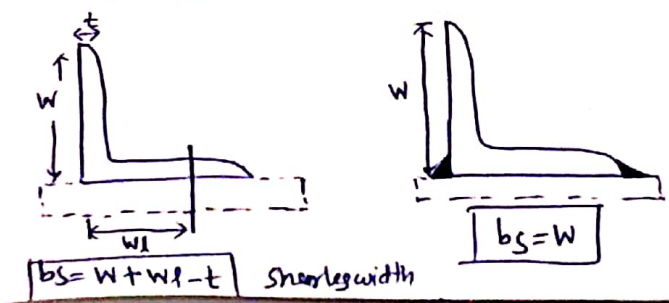
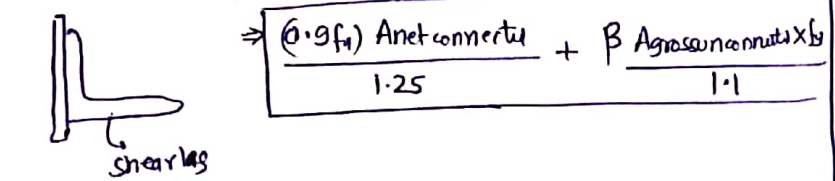
② cables → used in suspension bridge.  
 • generally bng, flexural stiffness → negligible  
 • initial sag & other geometrical effect must be accounted in the design.

③ Rods & bars :- used for small tension members  
 • such members in general welded to gusset plate or may be threaded and bolted

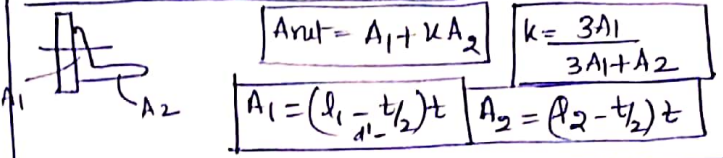
④ plates & flats :- used as tension member in -  
 transmission tower, foot bridge.  
 They are also used in column to keep the component member in their correct position like lacing flat, batten plates, end tie plate



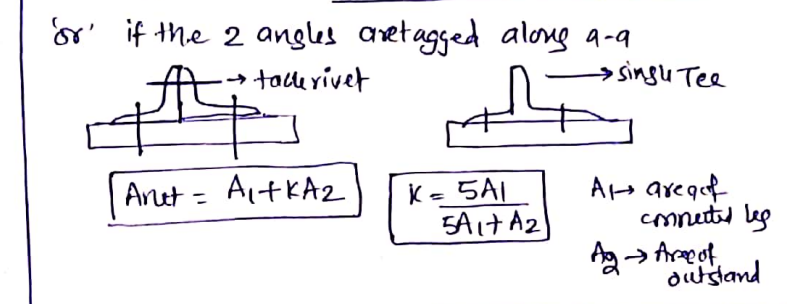
Rupture strength of Angle :- connected through 1 leg  
 is affected by shear lag.



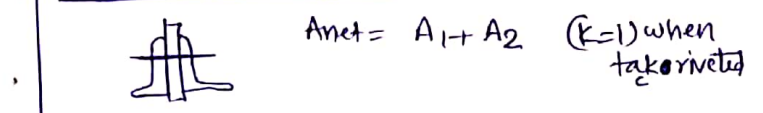
• single angle connected only 1 leg :-



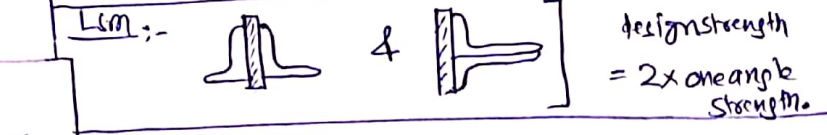
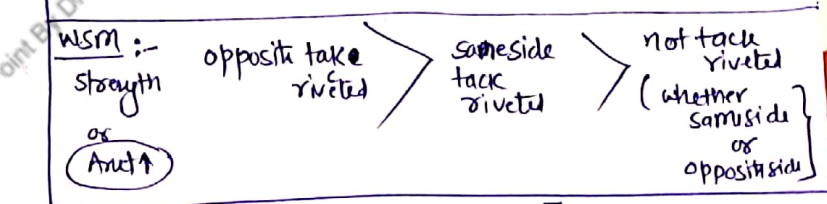
• pair of angle placed back to back (or a single Tee) connected by only one leg of each angle (or by the flange of tee) to the same side of gusset plate



• if 2 angles are placed back to back and connected to both side of gusset plate



if not tack riveted then both will be considered separately and use  $k = 3A_1 / (3A_1 + A_2)$



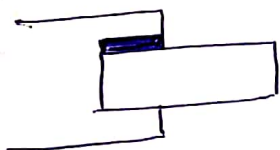
| Type of member                                                | $\lambda_{max}$<br>(Slenderness ratio) |
|---------------------------------------------------------------|----------------------------------------|
| Lacing bar in compression                                     | 145                                    |
| Bracing member in case of Hanger                              | 160                                    |
| always compressive (DL+LL)                                    | 180                                    |
| compressive (WL+EQ)                                           | 250                                    |
| tension member (load other than EQ)                           | 180                                    |
| tension member (WL+EQ)                                        | 350                                    |
| compression flange of beam against lateral torsional buckling | 300                                    |
| member always in tension except in pretensioned member        | 400                                    |

note: Tension member (Bracing) is Pretensioned to avoid sag  
 • need not to satisfy max. slenderness ratio

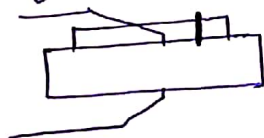
Lug angle :- ① reduce length of connection (save gusset plate)  
 ② reduce shear lag {  $\therefore \eta$  of Tension member  $\uparrow \uparrow \uparrow$  }  
 Thus stress-strain uniform hence no shear lag.

Isade specifications : (Lug angle):

① effective connection of lug angle shall be as far as possible Terminate at end of member



② fastening of Lug angle to main member :- shall preferably start in advance of direct connection of the member to the gusset or other supporting member.



③ lg + gusset plate  $\Rightarrow$  (min 2 bolts)

|                           | main member                      |
|---------------------------|----------------------------------|
|                           | angle channel                    |
| Lug angle to gusset plate | $+20\% \times F_{out\ standing}$ |
| Lug angle to main member  | $+40\% \times F_{out\ standing}$ |

Shear lag :- when stress in one part lags behind the other part of section.

- connected leg will have highest stress at failure than outstanding leg
- Shear lag reduce the effectiveness of component - plates of tension member that are not connected directly to a gusset plate.

note (for this reason unequal angle with long leg - connected is preferred)

Tension splice :- when 2 tension members are connected use it.

- design strength of tension splice  $\left\{ \begin{array}{l} \text{Yield} \\ \text{Rupture} \\ \text{Block shear} \end{array} \right\}$  min.
- splice connection design force  $\Rightarrow$  max  $\left\{ \begin{array}{l} 0.30 \times \text{member design capacity in Tension} \\ \text{designation (factored Load)} \end{array} \right.$   
 $\left\{ \begin{array}{l} \text{gross net Block} \\ \text{(yield)} \end{array} \right.$

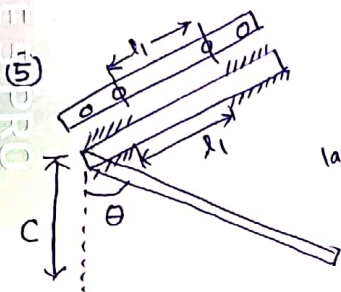




## Lacing Built up column :-

- ①  $\lambda_{max} = 14.5 = \frac{l_{eff}}{r_{min}} \left\{ r_{min} = \frac{t}{\sqrt{12}} \right\}$
- ②  $l_{eff} = 1.05 \times L_{eff} \left\{ \text{increase by 5\%} \right\}$   
(to account for shear deformation due to unbalanced horizontal SF.)
- ③ design as slender compression member (truss member)

- ④ lacing member — Rolled section, tubes of equal strength  
ISF, ISA, ISLB



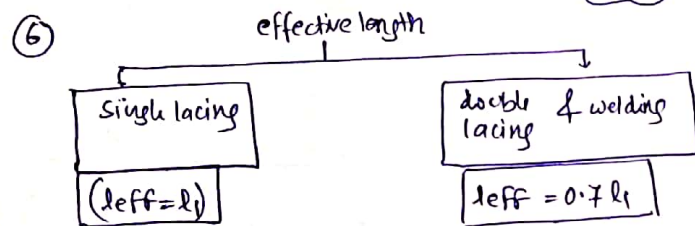
$$40 < \theta < 70$$

$$\frac{C}{r_{min}} \nless 50$$

$$\nless 0.7 \text{ whole}$$

$l_1 \Rightarrow$  inner to inner of bolt/weld

if fails then  
 $\theta \uparrow$   
provide double lacing



$$t_{min} < \begin{cases} \frac{l_1}{40} & \text{Single lacing \& welding} \\ \frac{l_1}{60} & \text{double lacing} \end{cases}$$

- ⑧ tie plates at end of lacing system  $\rightarrow$   
to prevent distortion of Built up c/s due to unbalanced horizontal force.

- ⑨ lacing should be designed to resist a transverse shear = 2.5% of axial column load (V)

$$\text{force in each lacing bar (F)} = \frac{V}{N \sin \theta} \begin{cases} N=2 \text{ Single} \\ N=4 \text{ double lacing} \end{cases}$$

$$\therefore \text{no of rivet req.} = \frac{2 F \cos \theta}{R_v}$$

- ⑩ lacing  $\rightarrow$  Best for eccentric loading  
if eccentric load then it is designed to resist additional shear caused due to BM.

$$\text{width of lacing bar} = 3 \times \text{dia of bolt}$$

| $\phi$ | B  |
|--------|----|
| 16     | 50 |
| 18     | 55 |
| 20     | 60 |
| 22     | 65 |

## Buckling of strut component (lock Bolted)

Trick T-24

$$\frac{l_1}{r_{min}} \nless 40$$

$$0.6 \text{ whole}$$

$$l_1 \rightarrow \text{dist. b/w lock bolts}$$

- Battened Built up column :-
- only axial load
  - not for eccentric load
  - design as frame

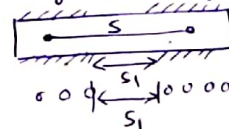
- Batten subjected to large SF & BM
- flat plate are used for Batten
- The no. of Battens  $\Rightarrow$  so that member divided into not less than 3 parts longitudinally  
 $\left\{ \begin{array}{l} \text{min. 4 batten plates or min 2 intermediate batten} \\ \& 2 \text{ two end battens} \end{array} \right\}$

- effective length of Battened column  $\Rightarrow +10\% \uparrow$   
increase by 10%.

- $V = 2.5\%$  of axial load

$$l_{min} \nless \frac{S_1}{50}$$

$S_1 \Rightarrow$  transverse dist b/w centroid of inner and bolt group or rivet group



$$F = \frac{VC}{NS}$$

$$M = F \times \frac{S_1}{2} = \frac{VC}{2N}$$

- effective depth of batten (d)  $\Rightarrow$  total depth of batten  $- 2 \times \text{edge dist}$

$$\begin{aligned} d &> \frac{3a}{4} && \rightarrow \text{for intermediate batten} \\ d &> a && \rightarrow \text{for end batten} \\ d &> 2b && \rightarrow \text{for any batten} \end{aligned}$$

## Design of slab base (IS 800:2007) :-

$$t = \sqrt{\frac{2.5 w (a^2 - 0.3 b^2)}{f_y / 1.1}}$$



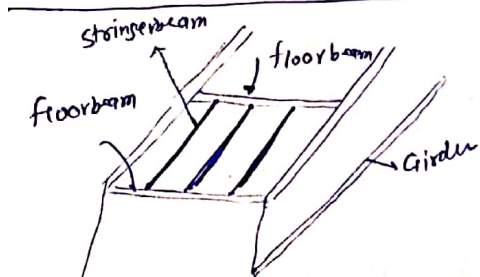
① filler joist :- • steel beam of light section  
Plain cement concrete.

[Joist → a beam supporting floor construction  
but not a major beam.]

② Girder :- floor beams used in Building  
↳ also major beam in any structure.

③ Floor beams :-  
↳ major beam supporting the beams  
↳ span b/w girder

④ Stringer beam → span b/w floor beam  
Stringers → members used in bridge parallel  
to the traffic to carry the deck slab • they  
will be connected by transverse floor beam



⑤ Spandrels :- exterior beam at floor level of building  
which carry part of floor load + exterior wall.

⑥ Lintel :- Beam members used to carry wall loads,  
over wall opening for door, window.

⑦ Purlins :- A roof beam usually supported  
by roof truss  
↳ supported on principal rafter

⑧ Rafter :- A roof beam usually supporting  
building.  
[rafter purlin roof truss]

⑨ Girt :- A horizontal members fastened to  
and spanning b/w peripheral column of an industrial  
building • used to support wall cladding like  
corrugated metal sheeting.

note: castellated beam :- (मकलबीम में Hole)



↳ wide flange I beam, I beam subjected to  
longitudinal cut along web.

Design criteria of beam :-

- 1- Design for BM
- 2- Design for SF
- 3- check for Deflection  $\left[ \frac{w_{sm}}{\text{max permissible deflection in SSB steel}} \right]$
- 4- check for secondary failure  $\neq \frac{\text{span}}{325}$

local buckling of  
compression flange  
or web

web crippling under  
concentrated load

Beam  
↳ laterally supported beam

whose compression  
flange restrain against  
lateral torsional buckling

↳ laterally unsupported beam

→ opposite

- ① Providing shear connector on compression flange.
- ② Bracing the compression flange of adjacent beam.

Bending (Flexural strength) :- (LSM)

$$\text{Design BM (M)} \leq \text{Design bending strength (M}_d\text{)}$$

(a) Laterally supported beam :-

when  $\frac{d}{t_w} < 67.6$  (no shear buckling in web)

Low Shear case

$$V_u \leq 0.6 V_d$$

$$\text{design shear strength} = \frac{A_v \times f_y}{\sqrt{3} \times 1.1}$$

↓  
Shear at web

web of section not susceptible to buckling under shear before yielding.

design bending strength of beam section (M<sub>d</sub>)

Plastic & compact section

$$M_d = \frac{(Z_p) f_y}{1.1}$$

Semi compact section

$$M_d = \frac{Z_e f_y}{1.1}$$

note:  
for SSB  $M_d < 1.2 \frac{Z_e f_y}{1.1}$   
for cantilever  $M_d < 1.5 \frac{Z_e f_y}{1.1}$

for slender section

$$M_d = Z_e f_y$$

↓  
reduced design strength

High Shear case

$$V_u > 0.6 V_d$$

Plastic or compact section -

$$M_{dv} = M_d - \beta (M_d - M_{dv}) \leq 1.2 \frac{Z_e f_y}{1.1}$$

$$\beta = \left( \frac{2V}{V_d} - 1 \right)^2$$

Semi compact

$$M_{dv} = \frac{Z_e f_y}{1.1}$$

Deflection limit :- Excessive deflection may lead to

Crack in plaster and may damage the material attached (or) supported by beam

IS 800:2007 :-

| vertical deflection for |                  | value                     |
|-------------------------|------------------|---------------------------|
| cantilever span         | elastic cladding | $\frac{\text{span}}{120}$ |
|                         | Brittle cladding | $\frac{\text{span}}{150}$ |
| Simply Purlin & girt    | elastic cladding | $\frac{\text{span}}{150}$ |
|                         | Brittle cladding | $\frac{\text{span}}{180}$ |
| Simply span             | elastic cladding | $\frac{\text{span}}{240}$ |
|                         | brittle "        | $\frac{\text{span}}{300}$ |

web crippling :- occurs at a point where concentrated load act.

Hence due to reaction at support, high compressive stresses are produced in web close to upper flange or lower flange



• web crippling is a bearing failure

• near the support web of beam may cripple due to lack of bearing capacity

note

• The crippling occurs at root of radius.



As per IS code  $\Rightarrow f_w = \frac{[b_1 + n_c] t_w}{1.1} f_y$   
to find web crippling of web  
(2.5 tf)

$n_c \rightarrow$  length obtain by dispersion through the flange to web at slope 1:2.5

note:-

How to avoid web ~~buckling~~ crippling

provide bearing stiffener

provide thicker webs

(b) laterally unsupported :- (refer IS code)



Web buckling :- • web buckling is the sudden sideways deflection of str. member under application of compressive load.



web buckling (vertical buckling) occurs when intensity of vertical compressive stress near centre of section becomes greater than the critical buckling stress for the web acting as a column. The buckling of column is much influenced by the restaint provided for the flanges.

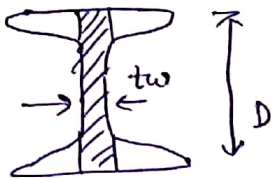
• web buckling is not a problem with-rolled beam section, this possibility exist in the thin webs of deep plate girders.

• normally if web is safe in crippling it will be safe in buckling also.

v.v. Imp conclusion :-

|                         |                                         |
|-------------------------|-----------------------------------------|
| ① Local Flange Buckling | → due to Bending compression            |
| ② Web crippling         | → more bearing stress at root of fillet |
| ③ Web buckling          | → diagonal compression due to shear     |

$$\text{Shear capacity of web} = \left[ \frac{f_y}{\sqrt{3} \times 1.1} \right] (D \times t_w)$$



Built up beam :-

① symmetrical built up beam

$$\text{Area of each cover plate} = \frac{Z - Z_1}{d}$$

$Z_1 \rightarrow$  section modulus of rolled I section available

② unsymmetrical built up beam

$$A_p = \frac{1.2(Z - Z_1)}{d}$$

note :-

Gross sectional area for flange plate is taken 20% more than the net cross-sectional area to allow for rivet holes and approximation in calculation.

Gantry girder :- [designed as laterally unsupported beam used in industrial building]

- subjected to unsymmetrical bending due to lateral Thrust.
- type of load acting over this

|                     |                                                          |
|---------------------|----------------------------------------------------------|
| ① Transverse load   | due to dead load (DL) → Gravity load                     |
| ② lateral load      | due to moving & stopping of crab                         |
| ③ longitudinal load | movement of truss of rails, starting & stopping of crane |

max. Deflection of Gantry girder under DL+LL (IS-800: 1984)

|                                         |                  |
|-----------------------------------------|------------------|
| where cranes → manually operated        | $\frac{L}{500}$  |
| EOT cranes < 50t                        | $\frac{L}{750}$  |
| EOT cranes > 50t                        | $\frac{L}{1000}$ |
| other moving loads such as charging car | $\frac{L}{600}$  |

### Seated connection :-

The connection b/w one beam to other beam or column with the angle at top & bottom.

### (i) unstiffened seated connection :-

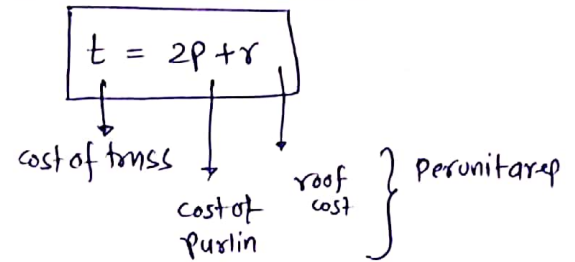
if packing stiffener is not provided with angle at top & bottom

(ii) stiffened seated connection :- if packing stiffener is provided with angle at top & bottom.

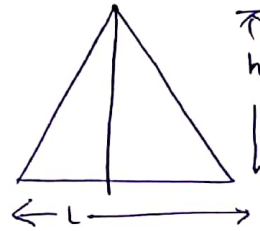
Purlin  $\rightarrow$  flexure member

Girts  $\rightarrow$  unsymmetrical bending

$$\text{max. BM in purlin} = \frac{wL^2}{10}$$



$$\text{economic spacing of truss} = \frac{L}{3} - \frac{L}{5}$$



$$\begin{aligned} \text{roof truss slope} \\ = 2 \times \text{pitch} \left( \frac{h}{L} \right) \end{aligned}$$

ToThePoint By DhyanPall



## Plati Girder [for larger & heavy Gravity load]

riveted or Bolted plate girder

welded-plate girder

provide flater-plate section

5-15% less weight than riveted/bolted plate girder

Provide Angles

(unequal leg with longer leg horizontal)

### Purpose of stiffener in a plate girder:

- to prevent buckling of web plate

Vertical stiffener (stability stiffener) (transverse stiffener)

- increase buckling resistance of web against shear
- Intermediate vertical stiffeners are joggled.

horizontal / longitudinal stiffener

- increase buckling resistance of web against bending.
- provided when the depth of web is more & there is tendency of web buckling.
- It is provided when vertical stiffeners becomes too close and only thin-plates are available for web.
- provided in compression zone of web.

Bearing stiffener or end bearing stiffener or load stiffener

- used to transfer concentrated loads on the girder & heavy reactions at support to full depth of web
- provide straight
- design as column with the length of web 20 times the thickness of web on both side.
- It prevents the web from crushing & buckling sideways.
- It relieves the rivet connecting the flange angles & web, from vertical shear.

diagonal stiffeners

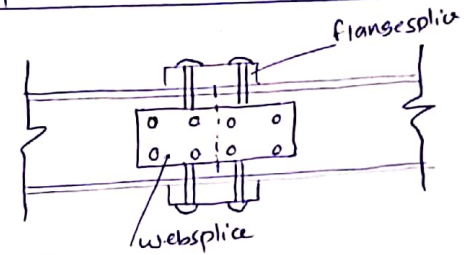
Safeweb against shear + bearing

Torsional stiffener

- To transmit Tensile forces applied to web through a flange
- provided at support to restrain the girder against Torsional effect.

### flangesplice:

- a joint in the flange element provided to increase the length of flange plate



- Flange splice is designed for axial force only.

### Web splice:- A joint in web plate to increase its length

- web splice designed for shear & moment at the spliced section
- the splice plates are provided on each side of web

$$\frac{d}{t_w} \leq 85$$

unstiffened  $\rightarrow$  does not require any stiffener

$$\frac{d}{t_w} > 85$$

provide transverse stiffener

$$85 < \frac{d}{t_w} < 200$$

only Intermediate vertical stiffener required.

$$200 < \frac{d}{t_w} < 250$$

vertical stiffener + 1 horizontal stiffener  
at a distance from compression flange equal to  $\frac{2}{5}$ th of the distance from the compression flange to the neutral axis are provided

$$250 < \frac{d}{t_w} < 400$$

vertical stiffener + 1 H.S. + 2nd H.S. at N.A.

- min distance b/w vertical stiffener  $\Rightarrow 0.33d$
- max  $\rightarrow 1.5d$

- min unsupported length of stiffener =  $1.80t_w$
- max =  $2.70t_w$

effective flange area in Compression =  $A_f + \frac{A_w}{6}$

Tension =  $A_f + \frac{3}{4} \left( \frac{A_w}{6} \right)$



Bernoulli  $\rightarrow$  strain diagram linear

$$Z_p = A_c y_c + A_t y_t$$

first choose equal arm axis

Plastic section modulus

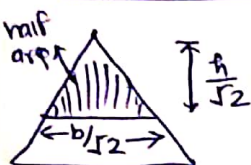
$$\text{Load factor} = f_o \underline{s} \times \underline{s} \text{ shape factor}$$

$$\frac{m_p}{m} = \frac{\text{Collapse load}}{\text{working load}}$$

$$\frac{m_y}{m}$$

$$\left(\frac{Z_p}{Z_e}\right) = \frac{m_p}{m_y}$$

$$\text{margin of safety} = f_o s - 1$$



$$\frac{d^2y}{dx^2} = \frac{1}{R} = \frac{m}{EI} = \frac{\sigma}{y} = \infty$$

$$\text{as } y \rightarrow 0$$

$\therefore$  Ratio of change of slope at plastic limit or curvature

Shape factor:  $(s)$



strong axis

$$(1.12 - 1.14)$$

$$\& (1.1 - 1.2)$$

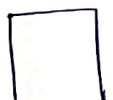
$$\text{weak } (1.50)$$



tubular

$$1.27$$

$$(4/\pi)$$



(rectangle)

$$1.50$$



wide flange

$$(weak axis)$$



$$1.70$$

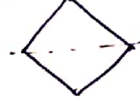
$$\text{or } 1.69$$

$$\&$$

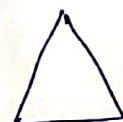
$$D/d = k$$

$$1.69 \left( \frac{1-k^3}{1-k^4} \right)$$

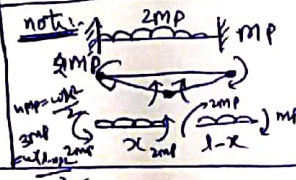
Rhombus



$$2$$



$$2.34$$



Design

elastic

equilibrium

yield condition

kinematic condition

compatibility condition

moment curvature relation

equilibrium

yield condition

Plastic

equilibrium

yield condition

kinematic condition

compatibility condition

moment curvature relation

equilibrium

yield condition

kinematic condition

Kinematic Theorem

Principle of Virtual Work

(mechanism)

$P \leq P_u$

$P = P_u$

Static Theorem

Lower Bound Theorem

$P \leq P_u$

$P = P_u$

$P = P_u$

Section

$\frac{b}{d}$

value

condition

class-1 (plastic)

$< 9.5$

$\frac{16MP}{12}$

Hinged develop, sufficient hinge to fail structure.

class-2 (compact)

$9.5 - 10.5$

$\frac{12MP}{12}$

Hinged develop, inadequate plastic hinge rotation capacity

class-3 (semi compact)

$10.5 - 15.7$

$m_y$

extreme fibre  $\sigma_f$   $\>$   $f_y$  due to local buckling can not develop MP

class-4 (slender)

$> 15.7$

$< m_y$

Before reaching  $m_y$ , element buckle locally

Plastic length  $L_p$  :- (length of elastoplastic zone)

$\rightarrow$  zone of yielding ( $m_p$  to  $m_y$ )

① SSB (conc. load mid)



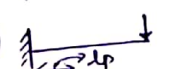
$$L_p = L \left( 1 - \frac{1}{5} \right)$$

② SSB (Udl)



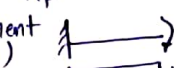
$$L_p = L \left[ 1 - \frac{1}{5} \right]$$

③ cantilever (conc. load free end)



$$L_p = L \left( 1 - \frac{1}{5} \right)$$

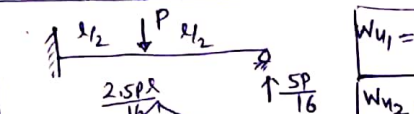
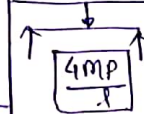
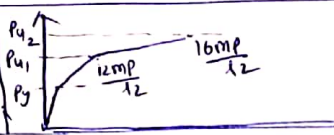
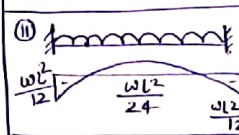
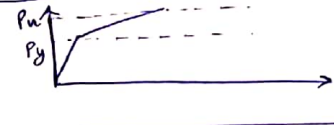
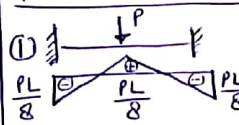
④ cantilever (conc. moment at end)



$$L_p = L$$

$L_p$  depend  $\rightarrow$  loading (conc. load or Udl)  
 $\rightarrow$  Geometry (shape factor  $s$ )  
 $\rightarrow$  length of beam

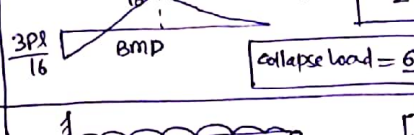
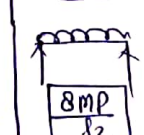
load Deflection diagram:



$$W_{u1} = \frac{16}{3} \frac{MP}{12} = 5.33 \frac{MP}{12}$$

$$W_{u2} = \frac{6MP}{12}$$

collapse load = 6MP

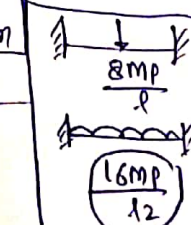


$$W_{u1} = 8MP/12$$

$$W_{u2} = 11.656MP/12$$

$\alpha = 0.4148 \rightarrow$  Plastic hinge form depend

collapse load = 11.656MP/12



noti:-

① no. of plastic hinge = degree of indeterminacy + 1

② no. of possible independent mechanism

$$= n - r \rightarrow \text{degree of freedom}$$

no. of possible plastic hinge

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